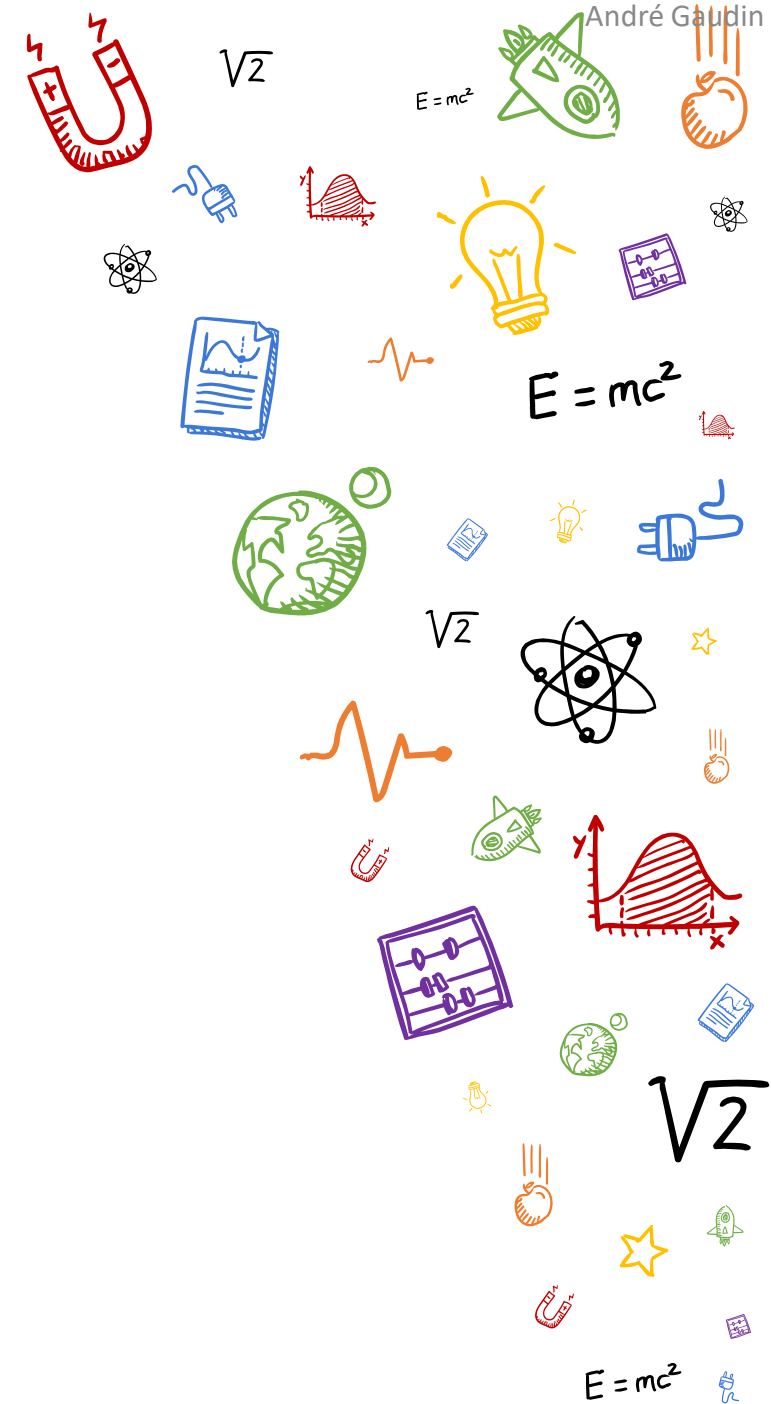


October 31st, 2023



Attendance Question



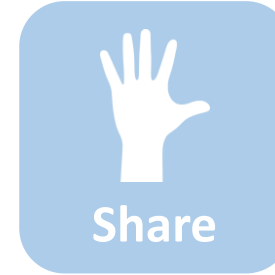
Want to ensure I know everyone's name. Please tell me your name (pronouns) and your answer:

Q: What is your **second favourite**
food?

Last Week's Lesson



What do you remember, about forces, from last week's lesson?



This Week and Today



This Week

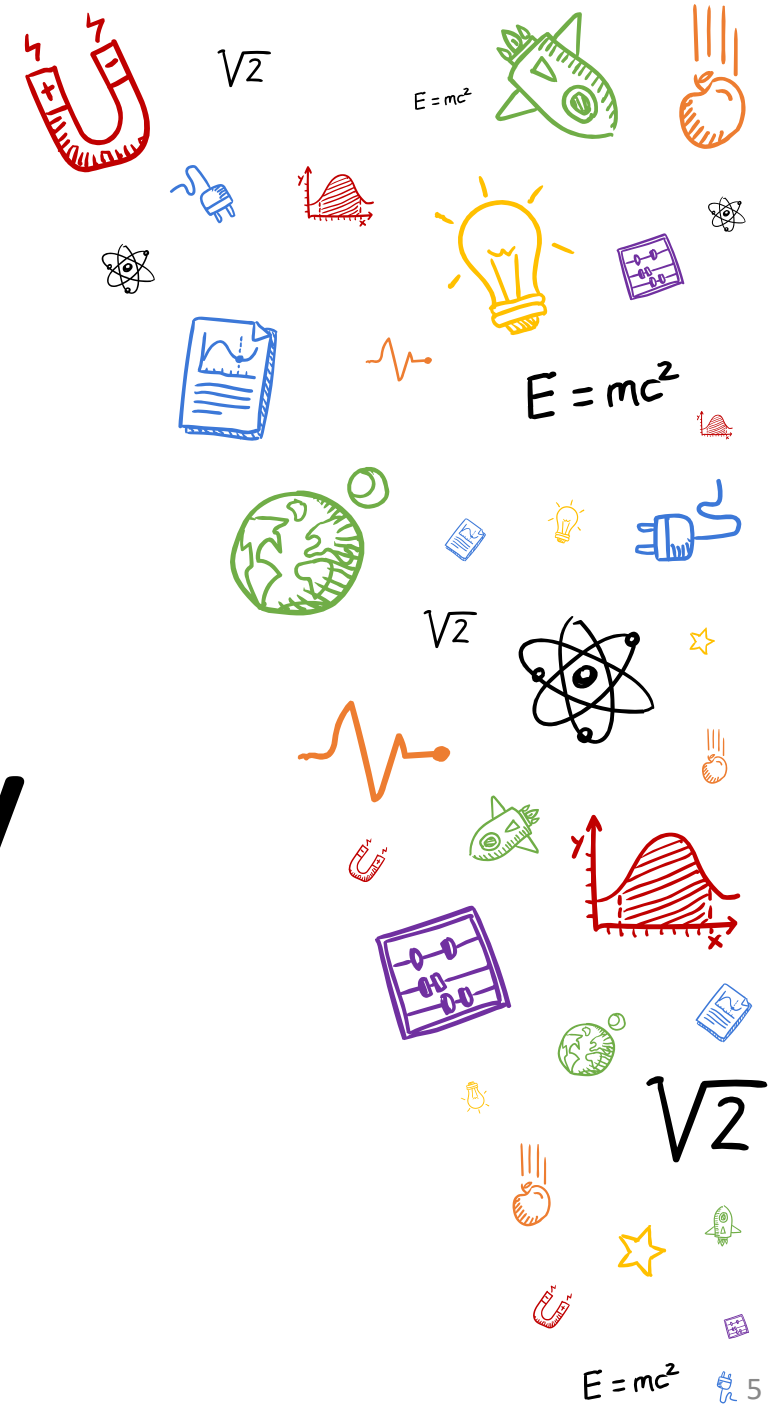
- Newton's Laws of Motion
 - First Law
 - Second Law
 - Third Law
- Surface Friction and Viscous Drag
- Elastic Restoring Forces and Hooke's Law
- Free Body Diagrams

Today

1. PhET Simulation of Motion
2. Newton's First and Second Law
3. Surface Friction



Starting Activity



PhET and Newton's Laws



Using a PhET simulation to explore the nature of motion and its connection to forces.

You will explore:

- a. What happens when no force is applied to a stationary? Describe the motion.
- b. What happens when no force is applied to a moving object? Describe the motion.



<https://phet.colorado.edu/en/simulations/forces-and-motion-basics>

 **PhET Forces and Motion: Basics**

PhET and Newton's Laws



Using a PhET simulation to explore the nature of motion and its connection to forces.

You will explore:

- c. What happens if a force is applied with to the direction of motion?
- d. What happens if a force is applied opposite to the direction of motion?



<https://phet.colorado.edu/en/simulations/forces-and-motion-basics>

 **PhET Forces and Motion: Basics**

Newton's First Law



Newton's first law of motion states that:

“An object remains **stationary** or moves at a **constant velocity** unless an **external force acts** on it.”

You might see it written with:

object or body

external force or **unbalanced force**

If $\sum F = 0$, then $v = \text{CONST}$

stationary: not moving ($v = 0$)

constant velocity: the speed isn't changing, nor the direction

external force: a force exerted by another object/body

unbalanced force: the net force is 0 ($\sum F = 0$)

Newton's First Law



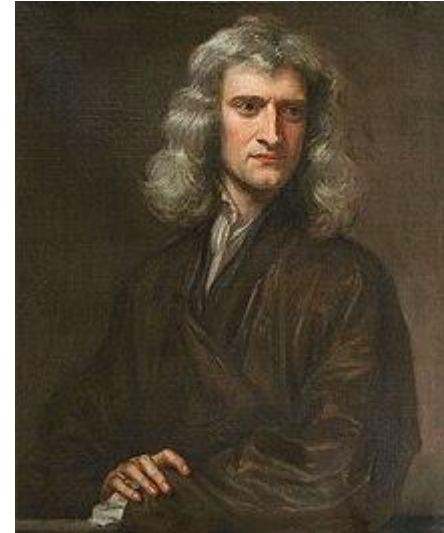
Newton's first law is based of his concept of net forces and Galileo's concept of inertia.

It says the **velocity** of an object **will not change** if there is **no net force** acting on it.

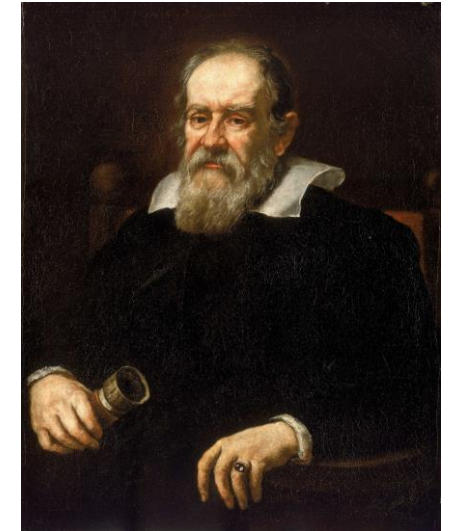
$$\text{If } \sum F = 0, \text{ then } v = \text{CONST}$$

Newton's wording:

*"Every body continues in its state of **rest**, or of **uniform motion in a straight line**, unless it is compelled to change that state by **forces** impressed thereon."*



Isaac Newton
(1643-1727)



Galileo Galilei
(1564-1642)

Newton's First Law: Question

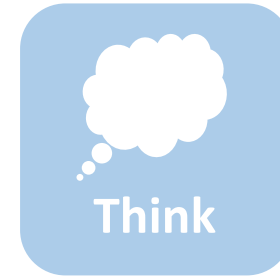


An object is observed to be moving at a constant speed but slowly changing direction, tracing out a large curve.

What can you say about the forces acting on the object?

Answer:

Since the direction is changing, that means the velocity is changing. From Newton's first law can say that the net force is not 0 N.



Newton's First Law: Question



A spaceship's is traveling through space far from any large bodies like stars or planets. A a certain point its rocket engine is turned off.

Describe the motion of the ship after the engine turns off.



Answer:

Since the engine turns off, and the spaceship is far from stars or planets, no force is acting on it. From Newton's first law, a net force of 0 N means the spaceship will continue moving in a straight line at a constant speed.

Newton's First Law



“What would your first law be given your everyday experience? Why?”

(“Using your everyday experiences only, how would the first law be different? Why?”)

An object will stop unless it is pushed on by a force.

~~FALSE~~



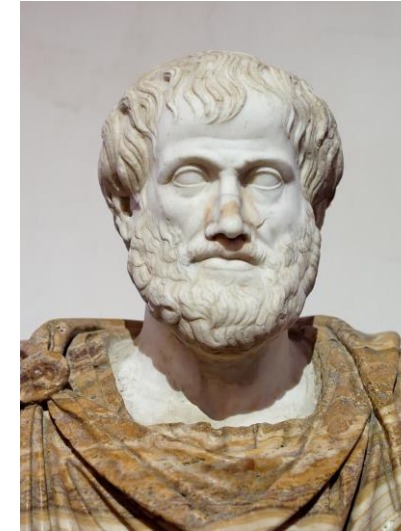
Think



Pair



Share



Aristotle
(384-322 BCE)

“The natural state of motion of all objects (but the heavenly ones) is one of rest.”

~~FALSE~~

Newton's Second Law



Newton's second law of motion states that:

The acceleration of an object is **directly proportional** to the net force action on the body and in the same direction:

$$F_{net} = ma \quad (\text{or } \sum F = ma)$$



$a = 0 \rightarrow \sum F = 0$ is the **condition for translational equilibrium.**



- **Linear relationship**
- m is a positive scalar multiplier, F and a have the same direction

Newton's Second Law: Question



An object is projected vertically upwards. Which is correct about the the net forces action on the object, at the instant when the object reaches its maximum height?

The net force:

- a. is 0 N.
- b. is a maximum.
- c. changes direction.
- d. is directed downward.

Explanation:

The force of gravity is always acting on the projectile, and it doesn't change direction or magnitude. From this, we know it is directed downward at the maximum, just like it is at all other points along the path.

Newton's Second Law: Question



A stationary tennis ball of mass 58 g is hit by a racket. The ball is in contact with the racket for a time of 10 ms and leaves the racket with a speed of 15 ms^{-1} .

Calculate the average force exerted by the racket on the ball.

We know:

$$u = 0 \text{ ms}^{-1}$$

$$v = 15 \text{ ms}^{-1}$$

$$t = 10 \text{ ms}$$

$$m = 58 \text{ g}$$

$$t = 10 \text{ ms} \times \frac{1 \text{ s}}{1000 \text{ ms}} = 0.010 \text{ s}$$

$$m = 58 \text{ g} \times \frac{1 \text{ kg}}{1000 \text{ g}} = 0.058 \text{ kg}$$

$$v = u + at$$

$$\begin{aligned} a &= \frac{v - u}{t} \\ &= \frac{15 \text{ ms}^{-1} - 0 \text{ ms}^{-1}}{0.010 \text{ s}} \\ &= 1500 \text{ ms}^{-2} \end{aligned}$$

Newton's Second Law: Question



A constant force of 4.0×10^2 N acts on an initially stationary railway car of mass 1.1×10^4 kg for 8.0 s.

Calculate, at 8.0 s:

- a. the velocity of the carriage,

$$F = ma$$

$$a = \frac{F}{m} = \frac{4.0 \times 10^2 \text{ N}}{1.1 \times 10^4 \text{ kg}} = 0.036 \text{ ms}^{-2}$$

$$v = u + at$$

$$\begin{aligned} &= 0 \text{ ms}^{-1} + 0.036 \text{ ms}^{-2} \cdot 8 \text{ s} \\ &= 0.29 \text{ ms}^{-1} \end{aligned}$$

- b. and the distance travelled.

$$s = vt + \frac{1}{2}at^2$$

$$= 0 \text{ ms}^{-1} \cdot 0 \text{ s} + \frac{1}{2} 0.036 \text{ ms}^{-2} \cdot (8 \text{ s})^2 = 1.2 \text{ m}$$

[illegible]

Surface/Solid Friction



You will **explore the force of friction** between some **wooden carts** (blocks) and **various surfaces**.



This may look simple, but you will have to approach it carefully to see each effect clearly and all the details of how the force changes.

Materials:

- 1 LabQuest 2 & 1 force sensor
- 1 wooden “cart”
- 1 each of: 1 kg and 500 g masses
- 1 tabletop (yes, that’s a surface too!)
- 1 sheet of mylar
- 1 sheet of plexiglass or 1 corkboard



The **mylar sheets** will be used later by the **Idea Lab** to make things, **please be careful** with them and **do not damage them** in any way.

Surface/Solid Friction



Goals:

1. What is the general shape of curve?
2. How does the nature of the surface affect the force of friction?
3. How does the mass of the “cart” affect the force of friction?

Materials:

- 1 LabQuest 2 & 1 force sensor
- 1 wooden “cart”
- 1 each of: 1 kg and 500 g masses
- 1 tabletop (yes, that’s a surface too!)
- 1 sheet of mylar
- 1 sheet of plexiglass or 1 corkboard

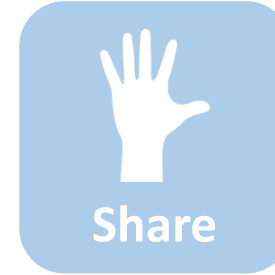


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[illegible]

Today's Lesson

- **What you learned today.**
- **What** questions you had before/concepts are **now clearer.**
- **What questions you still have.** (Write down in notebook.)



Wrap-up Question

“Which do you think is more powerful/useful and why, Newton’s first or second law?”

